

Project Report: Predicting Stock Price Movements Using Machine Learning

1. Objective

The goal of this project is to predict whether a selected stock's next day closing price will go up (1) or down (0) using historical stock data and technical indicators. The project demonstrates the ability to work with real-world financial data, build an end-to-end ML pipeline, and evaluate model performance effectively.

2. Implementation Summary

The notebook implements a complete machine learning workflow starting from data acquisition, feature engineering, and model training to evaluation and visualization. Each section is detailed below.

Data Acquisition

Stock data is fetched using the yfinance library for a two-year period.

The data includes standard OHLCV fields: Open, High, Low, Close, and Volume. It is chronologically ordered for time-series consistency.

Feature Engineering

Several technical indicators are created to capture trend, momentum, and volatility characteristics of the stock. These include Simple Moving Averages (SMA-10 and SMA-50), Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Bollinger Bands, and daily returns. Missing values from rolling windows are dropped to ensure data quality.

Target Variable

The target column 'Target' is defined as 1 if the next day's closing price is greater than today's, else 0. This is implemented by shifting the Close column and comparing the shifted values.

Data Splitting and Scaling

Data is split into training (80%) and testing (20%) sets based on time order to prevent data leakage. Features are standardized using StandardScaler for model stability.

Model Training

Three machine learning models are trained and compared: Logistic Regression, Random Forest, and XGBoost. Default or moderate hyperparameters are used to balance accuracy and generalization. Each model is trained on the processed features.

Model Evaluation

Performance metrics include Accuracy, Precision, Recall, F1-score, and ROC-AUC. Confusion matrices and ROC curves are plotted for each model to visualize performance. Random Forest and XGBoost outperform Logistic Regression.

Feature Importance and Insights

Feature importance plots highlight the most influential technical indicators. Momentum indicators such as RSI and MACD are found to be strong predictors, followed by moving averages. Correlation heatmaps are used to identify redundancy among features.

Predictions vs Actual Visualization

Predicted vs actual stock price directions are visualized to understand when the model performs well or fails. The plots indicate general trends and transitions between bullish and bearish phases.

Conclusions

- 1. XGBoost and Random Forest achieve better accuracy (around 0.60–0.65) than Logistic Regression.
- 2. The project demonstrates solid predictive modeling techniques for short-term stock movement prediction.
- 3. Future work could include integrating fundamental data, tuning hyperparameters, and using sequence models like LSTMs.

3. Overall Pipeline Summary

Pipeline Stage	Description	Tools / Libraries Used
Data Acquisition	Collected 2 years of historical OHLCV (Open, High, Low, Close, Volume) data for selected stocks.	yfinance
Feature Engineering	Computed key technical indicators such as RSI, SMA, MACD, Bollinger Bands, and daily Returns.	pandas, numpy
Labeling	Generated binary target labels for next-day direction (Up/Down) using shifted Close prices.	pandas.shift()
Data Splitting	Applied time-based split (80% train, 20% test) to preserve chronological order.	scikit-learn
Model Training	Trained multiple models for comparison — Logistic Regression, Random Forest, and XGBoost.	scikit-learn, xgboost
Evaluation Metrics	Evaluated model performance using Accuracy, F1 Score, ROC Curve, and Confusion Matrix.	scikit-learn.metrics, seaborn, matplotlib
Visualization	Created interactive and static plots for trends, ROC curves, and heatmaps.	matplotlib, seaborn

4.Results:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.4842	0.4444	0.2609	0.3288
Random Forest	0.5579	0.5345	0.6739	0.5962
XGBoost	0.6	0.5833	0.6087	0.5957

The observed model performances reflect the inherent challenges of predicting short-term stock movements. Logistic Regression achieved an accuracy of around 48%, indicating limited ability to capture the non-linear relationships in financial data. Ensemble models like Random Forest ($\approx 56\%$) and XGBoost ($\approx 60\%$) performed better, as they can model complex interactions among technical indicators such as RSI, SMA, and returns.

However, the moderate accuracy levels highlight the noisy and unpredictable nature of stock prices and the limited feature set used. The results represent statistical prediction quality rather than trading performance. Further improvements could be achieved through richer features, time-series cross-validation, and hyperparameter tuning to enhance model robustness and predictive reliability.